KIMBERLY ATTA-PETERS

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SUMMARY

A graduate student in Artificial Intelligence with experience developing machine learning models, data pipelines and AI applications using Python, SQL, AWS and modern MLOps practices. Experienced in feature engineering, model evaluation, experimentation and building data-driven solutions to support decision making.

EDUCATION

University of St. Thomas, MN — **May 2026 - Dec 2026**

Master of Science, Artificial Intelligence

Wisconsin International University of College, Ghana — **Aug 2014 - May 2018**

Bachelor of Laws

University of Ghana — **Aug 2012 - May 2016**

Bachelor of Arts, Information Systems, Psychology

WORK EXPERIENCE

University of St. Thomas

Graduate Research Assistant, Undergraduate Research Opportunities Program — **May 2025 - Present**

- Use Python and machine learning models to analyze research datasets and evaluate metrics such as accuracy, precision, recall and F1-score to identify patterns and support effective decision making
- Build dashboards and reports with Python, pandas, Power BI and Tableau to track participation, funding allocation and program trends
- Develop automated workflows using Power Automate, SharePoint and Salesforce to streamline data collection and application processing
- Analyze current and historical data to generate reports for the research board

Graduate Research Assistant, International Admissions — **Jun 2024 - May 2025**

- Developed SQL queries to analyze admissions data and identified trends that helped admissions counselors increase enrollment rates
- Created reports and visualizations in Python, Excel and pandas to communicate findings to support admissions decisions
- Managed student records and processed application documents using SharePoint, Salesforce and Ellucian Banner

Tuspa Company Limited

Data Analyst — **Feb 2023 - Apr 2024**

- Analyzed real estate datasets using Excel, Python, pandas and NumPy to identify trends and support business decisions
- Created weekly and monthly reports on occupancy, rental performance and expenses for business stakeholders
- Built dashboards in Power BI to track key metrics such as rent collection and property performance performance to support reporting need
- Cleaned, validated and standardized data to improve data quality and reporting accuracy

PROJECTS

AI Compliance Assistant (RAG-based System)

- Built a modular document processing pipeline with Python, PyMuPDF and pdfplumber to extract, clean and preprocess regulations with Git/Github for version control
- Developed a Retrieval Augmented Generation (RAG) system using Sentence Transformers, FAISS and large language models to perform regulatory compliance reviews and generate source-grounded recommendations.
- Designed a compliance scoring and review module to assess risk levels, identify missing requirements, flag violations and generate recommendations
- Developed RESTful APIs using FastAPI and orchestrated multi-stage document review workflows with LangChain and LangGraph, including document ingestion, retrieval, risk assessment, report generation and escalation routing
- Built the application with React and TypeScript, integrated automated reporting capabilities and containerized services with Docker for deployment on AWS

Asynchronous AI Inference System (MLOps Project)

- Built an end-to-end asynchronous machine learning inference system using Python, Apache Airflow, AWS S3, AWS SQS, Docker, and Kubernetes
- Developed automated Airflow workflows to train machine learning models, package artifacts, and store them in AWS S3 for scalable deployment.
- Designed a distributed inference architecture using AWS SQS and Kubernetes-based consumers to process prediction requests asynchronously
- Containerized services with Docker, deployed workloads on Kubernetes, and implemented data pipelines to support scalable, fault-tolerant inference and cloud-based storage of prediction outputs

Chicago Crime Prediction using Machine Learning

- Developed predictive models using Logistic Regression, Random Forest, XGBoost, Support Vector Machines, and Neural Networks to forecast crime occurrence using historical Chicago crime data.
- Engineered temporal and geographic features, including lag variables and rolling averages, to improve predictive performance
- Applied data preprocessing techniques, class imbalance handling, and model evaluation metrics to compare model effectiveness.
- Evaluated model performance using precision, recall, F1-score and ROC-AUC, identifying XGBoost and Logistic Regression as the strongest performing models

SKILLS

- **Programming Languages**: Python, SQL
- **Machine Learning Frameworks & Concepts**: TensorFlow, PyTorch, Scikit-learn, Keras, Machine Learning, Deep Learning, Neural Networks
- **AI, NLP**: NLP, LLM, RAG, Langchain, LangGraph
- **MLOPs**: Apache Airflow, Docker, Kubernetes, AWS S3, AWS SQS, FastAPI, Git/Github
- **Data Analysis & Visualization:** , Pandas, NumPy, Power BI, Tableau, Excel